

Using a Bomb Calorimeter to Determine the Enthalpy of Combustion for Naphthalene

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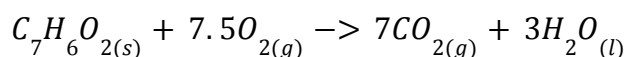
PURPOSE

The purpose of this investigation is to use a Parr oxygen bomb calorimeter to determine the standard molar enthalpy of combustion of naphthalene ($C_{10}H_8$). The heat capacity of the calorimeter was first determined by burning benzoic acid as a standard with a known enthalpy of combustion. This calibration constant was then applied to the combustion of naphthalene to find its enthalpy measured from the rise in temperature after the reaction. The experimental results were then compared to the literature value to assess accuracy, and Maxima was used to propagate the error and calculate the uncertainty in the calorimeter constant.

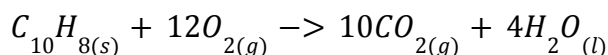
THEORY

The standard enthalpy of combustion, or ΔH_c° , is the net enthalpy change of a combustion reaction of a substance with oxygen to produce $CO_{2(g)}$ and $H_2O_{(l)}$ under standard temperature (25°C). The combustion reactions that are present in this investigation are:

Equation 1: Reaction - Benzoic Acid



Equation 2: Reaction - Naphthalene



Since the bomb calorimeter is a fixed volume, the combustion will occur under isochoric conditions. This means no pressure-volume work will be done, and the heat released will be ΔU as by the first law of thermodynamics:

Equation 3:

$$\Delta U = \Delta Q - w, w = 0 \text{ so } \Delta U = \Delta Q$$

The temperature change from T_i to T_f is proportional to the heat released, where C is the heat capacity of the entire system:

Equation 4:

$$Q_{cal} = C(T_f - T_i) = -\Delta U$$

The internal energy of a system (ΔU) can be rewritten to solve for the enthalpy (ΔH):

Equation 5:

$$\Delta H_c = \Delta U_c + \Delta n^G RT$$

R is the ideal gas constant ($8.314 \frac{J}{mol \cdot K}$), Δn^G is the change in moles of gas per mole of fuel, and T is the average of T_i and T_f in Kelvin. The calorimeter constant C is determined using Equation 6 below from the benzoic acid calibration trials. A correction factor must be added due to the heat released by the ignition of the wire that combusts during the experiment.

Equation 6:

$$C = (-n\Delta U_c + q_{wire})/\Delta T$$

EXPERIMENTAL

Equipment: Parr oxygen bomb calorimeter, pellet press, analytical balance (± 0.0001 g), thermometer ($\pm 0.01^\circ\text{C}$), oxygen cylinder, 2 L volumetric flask, ignition unit, and nichrome wire (2.3 cal/cm).

Approximately 1g of benzoic acid was weighed and transferred into the pellet press, forming a pellet. The mass of the pellet was recorded. A 10 cm length of nichrome wire was inserted into the top of the bomb head through the electrode holes and bent to make contact with the top of the pellet without touching the metal. The bomb was assembled, connected to an oxygen line, and purged for 30 seconds. It was then pressurized to 25 atm. The oxygen line was disconnected, and exactly 2.00 L of water was added to the calorimeter bucket. The bomb was lowered in, and the calorimeter was sealed with a stirring device in place. The temperature was recorded until the equilibrium was reached, then the ignition unit was plugged in, and the bomb was fired. The temperature was recorded each minute until it stabilized, after the bomb was vented and opened

to retrieve the remaining wire, which was then measured. The procedure was then repeated for the naphthalene trials.

DATA

The experimental data for all four trials conducted are reported below in **Table 1** and **Table 2**. T_i is the temperature recorded after the system equilibrated before ignition, and T_f is the middle value of the stable plateau region after combustion. The length of the wire burned was calculated by measuring the remaining wire present after the combustion and subtracting it from the initial 10.0cm, 2.3cal/cm was the manufacturer's value used to calculate q_{wire} .

Table 1: Summary of Experimental Data

Parameter	Benzoic Acid Trial 1	Benzoic Acid Trial 2	Naphthalene Trial 1	Naphthalene Trial 2
Mass of Pellet (g)	0.9693	1.0242	0.9838	1.0013
Wire Initial (cm)	10.0	10.0	10.0	10.0
Wire Final (cm)	5.40	5.35	4.50	4.20
Wire Burned (cm)	4.60	4.65	5.50	5.80
q_{wire} (J)	44.27	44.75	52.93	55.82
T_i (°C)	19.91	20.30	20.14	23.73
T_f (°C)	22.47	22.96	24.01	23.73
ΔT (°C)	2.56	2.66	3.87	3.97

Table 2: Temperature Over Time of Each Trial

Time (min)	Benzoic Acid Trial 1 (°C)	Benzoic Acid Trial 2 (°C)	Naphthalene Trial 1 (°C)	Naphthalene Trial 2 (°C)
0	19.91	20.30	20.14	19.76
1	21.10	22.07	21.46	21.60
2	21.89	22.55	23.05	22.96
3	22.24	22.82	23.69	23.44
4	22.38	22.93	23.91	23.62
5	22.44	22.95	23.98	23.69
6	22.46	22.96	24.01	23.72
7	22.47*	22.97	24.01*	23.73*
8	22.47	22.96*	24.01	23.73

* symbolizes the T_f values, while the T_i values were taken from $t = 0$.

CALCULATIONS

The standard molar internal energy of combustion for benzoic acid was calculated using the literature enthalpy value ($\Delta H_c = -3227.0 \pm 0.20 \text{ kJ/mol}$). Calculating Δn^G using **Equation 1** - **Reaction 1** yields $7 - 7.5 = -0.5 \text{ mol}$ of gas per mol benzoic acid. Trial 1 is used as an example below with $T = 294.340 \text{ K}$:

$$\Delta U_c = \Delta H_c - \Delta n^G RT$$

$$\Delta U_c = -3226700 - (-0.5)(8.314)(294.340)$$

$$\Delta U_c = -3227923 \text{ J/mol} = -3227.9 \text{ kJ/mol}$$

The Calorimeter constant C was then computed using **Equation 6**, once again using Trial 1 as an example:

$$C = \frac{(-(\frac{\text{mass of pellet}}{\text{molar mass of Benzoic Acid}})\Delta U_c + q_{\text{wire}})}{\Delta T}$$

$$C = \frac{(-(\frac{0.9693 \text{ g}}{122.12 \text{ g/mol}})(-3227923) + (44.27))}{2.56}$$

$$C = 10025.5 \text{ J/}^\circ\text{C}$$

The same calculation was performed using Trial 2, which yielded $C = 10186.7 \text{ J/}^\circ\text{C}$. The average of the calorimeter constant was taken and used later in the naphthalene calculations:

$$C = \frac{10025.5 + 10186.7}{2} = 10106.3 \text{ J/}^\circ\text{C}$$

The propagated error was determined using Maxima with the values $\delta m = \pm 0.0001 \text{ g}$ and $\delta T = \pm 0.01^\circ\text{C}$. For the full Maxima file, see the supplemental information.

$$\text{BA Trial 1: } C = 10025.46 \pm 39.18$$

$$\text{BA Trial 2: } C = 10194.29 \pm 38.34$$

With $C = 10106.3 \text{ J/}^\circ\text{C}$, the standard molar internal energy of combustion for naphthalene was found by rearranging **Equation 6**. Using Trial 1 as an example with

$$n = \frac{0.9838}{128.17} = 0.007676 \text{ mol and } T = 295.225 \text{ K:}$$

$$\Delta U_c = - \frac{(C\Delta T - q_{\text{wire}})}{n}$$

$$\Delta U_c = \frac{-((10106.3)(3.87) - 52.93)}{0.007676}$$

$$\Delta U_c = -5088386 \text{ J/mol} = -5088.3 \text{ kJ/mol}$$

Using **Equation 5**, ΔH_c was calculated with $\Delta n^G = 10 - 12 = -2 \text{ mol}$ of gas per mol naphthalene:

$$\Delta H_c = \Delta U_c + \Delta n^G RT$$

$$\Delta H_c = -5088386 + (-2)(8.314)(295.225)$$

$$\Delta H_c = -5093295 \text{ J/mol} = -5093.3 \text{ kJ/mol}$$

The same calculations were applied to Trial 2, yielding $\Delta H_c = -5133.7 \text{ kJ/mol}$.

The average of the experimental ΔH_c values was computed as the final result:

$$\Delta H_c^{\text{avg}} = \frac{(-5093.3 + -5133.7)}{2} = -5113.5 \text{ kJ/mol}$$

Comparing this to the literature value of -5150.09 kJ/mol gives a percent error of:

$$\% \text{ error} = \left| \frac{(-5113.5 - (-5150.09))}{-5150.09} \right| * 100 = 0.71\%$$

RESULTS

From this investigation the experimental standard molar enthalpy of combustion of naphthalene was determined to be -5113.5 kJ/mol. When compared to the literature value of -5150.09 kJ/mol it yielded a percent error of 0.71%. The two individual trial values of -5093.3 and -5133.7 kJ/mol were both less negative than the literature value, with Trial 2 coming closer at a difference of 16.4 kJ/mol compared to Trial 1 at 56.8 kJ/mol. The difference between the two naphthalene trials was 40.4 kJ/mol, indicating good replicability between runs.

The calculated experimental value is slightly less negative than the literature value. This is likely due to the loss of heat to the surroundings in our experiment, which would explain why it deviates slightly from the literature value. If the calorimeter is not perfectly adiabatic, heat will escape from the walls of the calorimeter instead of being absorbed by the water, lowering the result. A secondary source of error could be the wire correction, as there was no uncertainty associated with the measurement or the manufacturer value of 2.3 cal/cm with the nichrome wire. This would provide an overestimate of the combustion energy rather than an underestimate so it is likely a very minor source of error.

REFERENCES

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<https://webbook.nist.gov/cgi/cbook.cgi?ID=C65850&Mask=2>
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SUPPLEMENTAL INFORMATION

```

→ /* Constants */;
(%i1) M_BA : 122.12;
M_BA 122.12
(%i2) dUc : -3227923;
dUc -3227923
→ /* Benzoic acid trial data */;
(%i3) masses : [0.9693, 1.0242];
masses [0.9693, 1.0242]
(%i4) dTs : [2.56, 2.66];
dTs [2.56, 2.66]
(%i5) qwires : [44.27, 44.75];
qwires [44.27, 44.75]
→ /* Uncertainties */;
(%i6) dm : 0.0001;
dm 1.0 10-4
(%i7) ddT : 0.01;
ddT 0.01
→ /* Define C symbolically */;
(%i8) C_expr : (-(m/M_BA)·dUc + qw) / dT;
C_expr 
$$\frac{qw + 26432.386177530298 \text{ m}}{dT}$$

→ /* Partial derivatives */;
(%i9) dCdm_expr : diff(C_expr, m);
dCdm_expr 
$$\frac{26432.386177530298}{dT}$$

(%i10) dCddT_expr : diff(C_expr, dT);
dCddT_expr 
$$-\left(\frac{qw + 26432.386177530298 \text{ m}}{dT^2}\right)$$

→ /* Loop over BA trials */;

```



```
(%i11) for i : 1 thru 2 do (
    m_i : float(masses[i]),
    dT_i : float(dTs[i]),
    qw_i : float(qwires[i]),

    C_i : float(ev(C_expr, m=m_i, dT=dT_i, qw=qw_i)),

    sigma_C : float(sqrt(
        (float(ev(dCdm_expr, m=m_i, dT=dT_i, qw=qw_i)) * dm)^2 +
        (float(ev(dCddT_expr, m=m_i, dT=dT_i, qw=qw_i)) * ddT)^2
    )),

    print("BA Trial", i, ": C =", C_i, "J/°C, σC = ±", sigma_C, "J/°C")
);
BA Trial 1 : C = 10025.461688234422 J/°C, σC = ± 39.17556861725935 J/°C
BA Trial 2 : C = 10194.285685348319 J/°C, σC = ± 38.337262734583184 J/°C
(%o11) done
```

Propagation of Error

Given an equation $y = f(a, b, c)$ such as $y = 3a + 2bc$, where σ_a , σ_b , σ_c are the errors in the measurements of variables a , b , and c , the question arises of how much error transfers to the quantity “ y ”.

It turns out that for a quantity “ y ” calculated from measured variables “ a , b , and c ” the error propagated to “ y ” symbolized by σ_y can be calculated as shown below:

$$\sigma_y = \sqrt{\left(\frac{\partial y}{\partial a}\right)^2 * \sigma_a^2 + \left(\frac{\partial y}{\partial b}\right)^2 * \sigma_b^2 + \left(\frac{\partial y}{\partial c}\right)^2 * \sigma_c^2 + \dots}$$

The heart of the calculation lies in taking the partial derivative of the function with respect to all measured variables. It assumes there is no correlation between the sources of error.

As an example let's assume we took pressure (P), volume (V), and temperature (T) measurements and these measurements are used to calculate a value of “ y ” as described in equation (2) below:

$$y = PV/T \quad (2)$$

The following are the derivatives of y with respect to each of the variables:

$$dy/dP = V/T, \quad dy/dV = P/T, \quad dy/dT = -PV/T^2$$

This gives for equation (1): $\sigma_y = ((V/T)^2 \sigma_P^2 + (P/T)^2 \sigma_V^2 + (-PV/T^2)^2 \sigma_T^2)^{1/2}$